

# THE PARITY CRITERION FOR NATIVE CHIRAL PROJECTORS: MIDDLE-GRADE HODGE DUALITY ON THE BIQUATERNIONIC CARRIER

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ORCID: 0009-0000-0133-2291. Standalone companion to *The Causal Accumulation Law: The Four Corners of  $\alpha$ , Algebraic Closure, and the Standard Model* [12], discharging the native-chiral-projector claim of that manuscript. All numerical claims refer to the harness distributed with this paper [15], `chiral_projector_harness.py`, which computes in exact integer and rational arithmetic, uses no tolerances, and exits with status equal to the number of disagreements. Dated July 27, 2026.

ABSTRACT. The framework's chiral projector  $P_{\pm} = \frac{1}{2}(1 \pm \star)$  is stated in the series as a native object, available on the carrier without insertion of an auxiliary complex structure. This note proves the exact condition under which that statement holds and exhibits the failure mode when it does not. The criterion is a parity condition and nothing else: on  $\Lambda^k$  of an  $n$ -dimensional space with a nondegenerate form of signature  $(p, q)$ , one has  $\star\star = +\text{id}$  if and only if  $k(n - k) + q$  is even. At the middle grade of an eight-dimensional carrier,  $k(n - k) = 16$  is even, so the condition reduces to the parity of the negative count: the projector exists precisely when  $q$  is even. The carrier signature is therefore not a modelling preference but the necessary and sufficient condition for real chirality at the middle grade. Three further results are proved and then measured. First, whenever the projector exists, the  $\pm 1$  eigenspaces have equal dimension, because  $\star$  has identically zero diagonal in a multi-index basis and hence vanishing trace; at  $n = 8$  this forces the split  $70 = 35 \oplus 35$ . Second, the middle grade is the unique grade at which the split is internal to a single grade, since for  $k \neq n - k$  the star is not an endomorphism of  $\Lambda^k$  at all; on the paired space  $\Lambda^k \oplus \Lambda^{n-k}$  the same parity criterion governs, so in an even- $q$  signature the projector is available at every even grade and at no odd grade, and in an odd- $q$  signature exactly the reverse. Third, the two block volume forms  $\omega_1 = e^0 \wedge \cdots \wedge e^3$  and  $\omega_2 = e^4 \wedge \cdots \wedge e^7$  span a two-dimensional  $\star$ -invariant subspace whose eigenvectors are  $\Psi_{\pm} = \omega_1 \pm \omega_2$  with eigenvalues  $\pm \varepsilon$ , where  $\varepsilon = (-1)^{q_1}$  is fixed by the number of negative directions in the first block; the handedness label is therefore convention-dependent and must be declared. When the parity condition fails, the failure is exact and scalar:  $P_{\pm}^2 - P_{\pm} = -\frac{1}{2}\text{id}$  and  $P_+P_- = +\frac{1}{2}\text{id}$ , and both  $P_{\pm}$  become invertible, so instead of a decomposition into halves one obtains two operators of full rank. Every statement is verified symbolically over the rationals on eleven signatures at  $n = 8$  and exhaustively for  $2 \leq n \leq 8$  across all grades and all signatures, with zero disagreements between the closed form and the computed matrix. The classical four-dimensional case is recovered as a corollary: self-duality is a real splitting in Euclidean and split signature and a complex structure in Lorentzian signature, by the same parity count.

A final section identifies what the criterion is and what it is not. The middle-grade star and the Clifford volume element have the same square identically, the exponents differing by  $m(1 - m)$ , so the criterion is not an analogue of the spinor chirality condition but the same condition on the same algebra element in a different module: it is the Weyl condition. It is therefore a mod-4 condition, while the real Clifford algebras have period 8, and the missing factor of two is not a classification label. It is the reality of the spinor module, that is, the Majorana condition, read off the signature of the trace form in the blade basis. The two conditions are separately mod 4 and mod 8 and their conjunction is  $p - q \equiv 0 \pmod{8}$ , which is the Majorana–Weyl condition and is where the period lives. This closes an item the earlier draft left open. Imposing real chirality, a real structure, and a nonempty light cone in eight dimensions leaves exactly one signature,  $(4, 4)$ : the two quaternionic candidates  $(6, 2)$  and  $(2, 6)$  carry only a symplectic Majorana condition, and the two definite candidates have no causal structure. The forcing selects the signature given the dimension and does not select the dimension, since  $n = 2, 4, 6, 8$  each admit exactly one indefinite Majorana–Weyl signature while  $n \geq 10$  admit several; the dimension comes from the division algebra tower, which is a separate input and should be stated as a separate link. Every statement in this section is verified on 189 signatures with  $n \leq 18$ , with the mod-8 dependence exhibited with zero conflicts, the type discriminator anchored on seven known algebras, and the Lorentzian Majorana–Weyl dimensions recovered as  $d \equiv 2 \pmod{8}$ .

## 1. SCOPE AND CLAIM

The series has asserted, without proof, that the carrier signature is what admits the native chiral projector  $P_{\pm} = \frac{1}{2}(1 \pm \star)$ , and that the even count of negative directions is what permits it. This note discharges that assertion. The claim proved here is the following.

Let  $V = \mathbb{R}^n$  carry a nondegenerate diagonal symmetric form of signature  $(p, q)$  and let  $\star$  denote the Hodge star of Convention 2.2. Then  $\star\star = (-1)^{k(n-k)}(-1)^q \text{id}$  on  $\Lambda^k$ , and consequently  $P_{\pm} = \frac{1}{2}(1 \pm \star)$  is a pair of complementary orthogonal

idempotents if and only if  $k(n - k) + q$  is even. At  $n = 8$ ,  $k = 4$  this is exactly the condition that  $q$  be even, and the eigenspaces then have dimension 35 each.

**Status bookkeeping**, in the manner of the series.

*Unconditional and proved:* the star-square identity (Lemma 3.1); the projector dichotomy with its exact failure constants (Theorem 3.2); the equal split by the trace argument (Proposition 4.1); the uniqueness of the middle grade and the paired-grade extension (Proposition 4.3); the block eigenvectors and the handedness label (Proposition 5.1); the four-dimensional corollary (Corollary 8.1); the identification of the criterion with the Weyl condition (Lemma 9.1); the Majorana / Weyl split of the mod-8 period (Proposition 9.5, Corollary 9.7); the signature forcing in eight dimensions (Theorem 9.10).

*Measured:* every proved statement above, recomputed symbolically over the rationals on eleven signatures at  $n = 8$  and exhaustively for  $2 \leq n \leq 8$  over all grades and all signatures, with zero disagreements (Section 7); and the Clifford-side statements on 189 signatures with  $n \leq 18$ , with the type discriminator anchored on seven known algebras and the mod-8 dependence exhibited with zero conflicts (Section 9).

*Carried, not proved here:* the identification of  $\star$  with the pseudoscalar of the even subalgebra and hence with the privileged central imaginary of the source current (Remark 8.2); the triality identification of the two 35-dimensional halves (Remark 9.13).

*Closed since the previous draft:* that draft recorded as open the question of what distinguishes  $(4, 4)$  from the other signatures satisfying the criterion, on the ground that the criterion selects a parity class rather than a signature. Theorem 9.10 settles it. The criterion is one of two conditions; the other is the Majorana condition; their conjunction plus a nonempty light cone leaves  $(4, 4)$  alone in eight dimensions.

*Open, stated:* whether the two 35-dimensional halves carry the representation content the framework requires, as against having the right dimension (Remark 10.3); and the fact that the forcing selects a signature and not a dimension (Remark 9.12).

## 2. CONVENTIONS AND DEFINITIONS

**Convention 2.1** (Carrier and form).  $V = \mathbb{R}^n$  with basis  $e_0, \dots, e_{n-1}$  and dual coframe  $e^0, \dots, e^{n-1}$ . The form is  $\eta = \text{diag}(\eta_0, \dots, \eta_{n-1})$  with  $\eta_i \in \{+1, -1\}$ ;  $q$  denotes the number of indices with  $\eta_i = -1$ , so  $\text{sgn det } \eta = (-1)^q$ . For a sorted multi-index  $I = (i_1 < \dots < i_k)$  write  $e^I = e^{i_1} \wedge \dots \wedge e^{i_k}$ ,  $\varepsilon_I = \prod_{i \in I} \eta_i$ , and  $I^c$  for the sorted complement. The  $\binom{n}{k}$  forms  $e^I$  are the standard basis of  $\Lambda^k$ .

**Convention 2.2** (Hodge star).  $\star : \Lambda^k \rightarrow \Lambda^{n-k}$  is the linear map

$$(1) \quad \star e^I = \varepsilon_I \text{sgn}(I, I^c) e^{I^c},$$

where  $\text{sgn}(I, I^c)$  is the sign of the permutation carrying  $(0, 1, \dots, n-1)$  to the concatenation  $(I, I^c)$ . This is the map fixed by  $\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_\eta \text{vol}$  on an orthonormal coframe, with  $\text{vol} = e^0 \wedge \dots \wedge e^{n-1}$ .

**Definition 2.3** (Chiral projectors). When  $n = 2m$  and  $k = m$ , so that  $\star$  is an endomorphism of  $\Lambda^m$ , set

$$(2) \quad P_\pm := \frac{1}{2}(\text{id} \pm \star) \in \text{End } \Lambda^m.$$

$P_\pm$  is called a *chiral pair* if  $P_\pm^2 = P_\pm$ ,  $P_+ + P_- = \text{id}$ , and  $P_+ P_- = P_- P_+ = 0$ .

**Definition 2.4** (Paired-grade operator). For  $k \neq n-k$  define  $Z_k \in \text{End}(\Lambda^k \oplus \Lambda^{n-k})$  by  $Z_k(\alpha, \beta) = (\star \beta, \star \alpha)$ , and  $P_\pm^{(k)} = \frac{1}{2}(\text{id} \pm Z_k)$ .

## 3. THE PARITY CRITERION

**Lemma 3.1** (Star square). *On  $\Lambda^k(V)$ , with Conventions 2.1 and 2.2,*

$$(3) \quad \star\star = (-1)^{k(n-k)} (-1)^q \text{id}.$$

*In particular  $\star\star = +\text{id}$  if and only if  $k(n-k) + q$  is even.*

*Proof.* Apply (1) twice:

$$\star\star e^I = \varepsilon_I \text{sgn}(I, I^c) \star e^{I^c} = \varepsilon_I \varepsilon_{I^c} \text{sgn}(I, I^c) \text{sgn}(I^c, I) e^I.$$

Two factors are to be evaluated. First,  $\varepsilon_I \varepsilon_{I^c} = \prod_{i=0}^{n-1} \eta_i = (-1)^q$ , since  $I$  and  $I^c$  partition the index set. Second,  $(I^c, I)$  is obtained from  $(I, I^c)$  by moving a block of  $n-k$  entries past a block of  $k$  entries, which is  $k(n-k)$  adjacent transpositions, so  $\text{sgn}(I, I^c) \text{sgn}(I^c, I) = (-1)^{k(n-k)}$ . Neither factor depends on  $I$ , so the composite is a scalar multiple of the identity with the stated scalar.  $\square$

**Theorem 3.2** (Parity criterion for the chiral pair). *Let  $n = 2m$  and let  $P_{\pm}$  be as in (2). Write  $s = (-1)^{m^2+q}$  for the scalar of Lemma 3.1 at  $k = m$ . Then the following are equivalent:*

- (i)  $m^2 + q$  is even;
- (ii)  $\star\star = +\text{id}$  on  $\Lambda^m$ ;
- (iii)  $P_+^2 = P_+$ ;
- (iv)  $P_+ P_- = 0$ ;
- (v)  $P_{\pm}$  is a chiral pair.

*If instead  $m^2 + q$  is odd, then exactly and with no error term,*

$$(4) \quad P_{\pm}^2 - P_{\pm} = -\frac{1}{2} \text{id}, \quad P_+ P_- = +\frac{1}{2} \text{id},$$

*$P_+ + P_- = \text{id}$  still holds, and  $P_{\pm}$  are invertible, so  $\text{rank } P_+ = \text{rank } P_- = \dim \Lambda^m$ .*

*Proof.* Lemma 3.1 gives  $\star\star = s \text{id}$  with  $s = (-1)^{m(n-m)+q} = (-1)^{m^2+q}$ , which is (i)  $\Leftrightarrow$  (ii). Expanding (2),

$$P_{\pm}^2 = \frac{1}{4}(\text{id} \pm 2\star + s \text{id}), \quad P_+ P_- = \frac{1}{4}(\text{id} - \star\star) = \frac{1-s}{4} \text{id}.$$

If  $s = +1$  then  $P_{\pm}^2 = \frac{1}{2}(\text{id} \pm \star) = P_{\pm}$  and  $P_+ P_- = 0$ , and  $P_+ + P_- = \text{id}$  is immediate from the definition, giving (v). If  $s = -1$  then  $P_{\pm}^2 = \pm \frac{1}{2} \star$ , whence  $P_{\pm}^2 - P_{\pm} = \pm \frac{1}{2} \star - \frac{1}{2} \text{id} \mp \frac{1}{2} \star = -\frac{1}{2} \text{id}$ , and  $P_+ P_- = \frac{1}{2} \text{id}$ . This proves (4) and shows that each of (iii), (iv), (v) fails when (ii) fails, completing the equivalence. For invertibility when  $s = -1$ :  $\star$  satisfies  $\star^2 = -\text{id}$ , so its eigenvalues are  $\pm i$  and those of  $\text{id} \pm \star$  are  $1 \pm i \neq 0$ ; equivalently  $(\text{id} \pm \star)(\text{id} \mp \star) = \text{id} - \star\star = 2 \text{id}$ , exhibiting the inverse.  $\square$

**Corollary 3.3** (The carrier condition). *At  $n = 8$ ,  $m = 4$  one has  $m^2 = 16$ , even, so Theorem 3.2 reduces to:  $P_{\pm}$  is a chiral pair on  $\Lambda^4(\mathbb{R}^8)$  if and only if  $q$  is even. The carrier signature of the series,  $(1, 3) \oplus (3, 1)$  with  $q = 4$ , satisfies this, and no signature with an odd number of negative directions does.*

*Remark 3.4.* Corollary 3.3 is the content of the series' claim that the signature is forced rather than chosen. It is worth stating what is *not* being claimed. The condition is on the parity of  $q$  alone, so  $(2, 6)$ ,  $(6, 2)$ ,  $(0, 8)$  and  $(8, 0)$  satisfy it as well; the criterion selects a parity class, not a unique signature. What singles out  $(4, 4)$  within that class must come from elsewhere in the framework, and does not come from this note.

## 4. THE SPLIT IS EQUAL, AND THE MIDDLE GRADE IS UNIQUE

**Proposition 4.1** (Equal split). *Let  $n = 2m$  with  $m \geq 1$  and suppose  $\star\star = +\text{id}$  on  $\Lambda^m$ . Then  $\text{tr } \star = 0$ , and consequently*

$$(5) \quad \text{rank } P_+ = \text{rank } P_- = \frac{1}{2} \binom{n}{m}.$$

At  $n = 8$  this is  $\text{rank } P_+ = \text{rank } P_- = 35$  and  $70 = 35 \oplus 35$ .

*Proof.* By (1) the star carries the basis element  $e^I$  to a scalar multiple of  $e^{I^c}$ . Since  $I$  and  $I^c$  are disjoint and both nonempty for  $1 \leq m \leq n-1$ , we have  $I \neq I^c$ , so every diagonal entry of the matrix of  $\star$  in this basis is zero and  $\text{tr } \star = 0$ . Under the hypothesis  $\star$  is an involution, so it is diagonalizable with eigenvalues in  $\{+1, -1\}$  and multiplicities  $d_{\pm}$  satisfying  $d_+ + d_- = \binom{n}{m}$  and  $d_+ - d_- = \text{tr } \star = 0$ . Finally  $P_{\pm}$  is the spectral projector onto the  $\pm 1$  eigenspace, so  $\text{rank } P_{\pm} = d_{\pm}$ .  $\square$

*Remark 4.2.* The trace argument is worth isolating because it makes the equal split independent of the signature, of the arrangement of the negative directions, and of any computation. The matrix of the star has no diagonal at all. Everything else about the operator can change and the balance cannot.

**Proposition 4.3** (Uniqueness of the middle grade, and the paired extension). *For  $k \neq n-k$  the star is not an endomorphism of  $\Lambda^k$ , and no operator of the form  $\frac{1}{2}(\text{id} \pm \star)$  is defined on  $\Lambda^k$ . On the paired space of Definition 2.4 one has*

$$(6) \quad Z_k^2 = (-1)^{k(n-k)+q} \text{id},$$

so  $P_{\pm}^{(k)}$  is a chiral pair on  $\Lambda^k \oplus \Lambda^{n-k}$  under exactly the same parity condition, with ranks  $\binom{n}{k}$  each. In particular, at  $n = 8$ :

- (a) if  $q$  is even,  $P_{\pm}^{(k)}$  is a chiral pair for every even  $k$  and for no odd  $k$ ;
- (b) if  $q$  is odd,  $P_{\pm}^{(k)}$  is a chiral pair for every odd  $k$  and for no even  $k$ ;
- (c) the middle grade  $k = 4$  is the unique grade at which the pair acts inside a single grade rather than on a sum of two.

*Proof.*  $Z_k^2(\alpha, \beta) = (\star\star\alpha, \star\star\beta)$ , and Lemma 3.1 applies on each summand with the same scalar, since  $k(n-k) = (n-k)k$ ; this is (6). Theorem 3.2 then applies verbatim with  $Z_k$  in place of  $\star$ . For  $n = 8$ ,  $k(8-k)$  is even exactly when  $k$  is even, which gives (a) and (b). Part (c) is the observation that  $\star$  maps  $\Lambda^k$  into  $\Lambda^{n-k}$ , so it is an endomorphism of a single grade only when  $k = n-k$ .  $\square$

*Remark 4.4.* Proposition 4.3(b) sharpens what the control run measures. In an odd- $q$  signature the chiral pair is not absent; it has moved to the odd grades. What is absent there is a chiral pair at the middle grade, which is the only place a pair can act without pairing two grades together. So the correct statement of the framework's requirement is not "chirality needs an even negative count" but "chirality internal to one grade needs an even negative count," and the two clauses are separately necessary: even  $k$  and even  $q$ , or odd  $k$  and odd  $q$ .

## 5. BLOCK VOLUME FORMS AND THE HANDEDNESS LABEL

Let  $n = 8$  and let the carrier be presented as a sum of two four-dimensional blocks, indices  $\{0, 1, 2, 3\}$  and  $\{4, 5, 6, 7\}$ , as the decomposition  $(1, 3) \oplus (3, 1)$  does. Write

$$(7) \quad \omega_1 = e^0 \wedge e^1 \wedge e^2 \wedge e^3, \quad \omega_2 = e^4 \wedge e^5 \wedge e^6 \wedge e^7,$$

the two block volume forms, and  $\Psi_{\pm} = \omega_1 \pm \omega_2$ .

**Proposition 5.1** (Block eigenvectors). *Let  $\varepsilon_1 = \prod_{i < 4} \eta_i$  and  $\varepsilon_2 = \prod_{i \geq 4} \eta_i$ , and let  $q_1$  be the number of negative directions in the first block. Then*

$$(8) \quad \star \omega_1 = \varepsilon_1 \omega_2, \quad \star \omega_2 = \varepsilon_2 \omega_1.$$

*If  $q$  is even then  $\varepsilon_1 = \varepsilon_2 =: \varepsilon = (-1)^{q_1}$ , the span of  $\omega_1, \omega_2$  is  $\star$ -invariant, and*

$$(9) \quad \star \Psi_{\pm} = \pm \varepsilon \Psi_{\pm}.$$

*If  $q$  is odd then  $\varepsilon_2 = -\varepsilon_1$ , the span is still invariant, and  $\Psi_{\pm}$  are not eigenvectors.*

*Proof.* For  $I = \{0, 1, 2, 3\}$  the concatenation  $(I, I^c) = (0, 1, \dots, 7)$  is the identity permutation, so  $\text{sgn}(I, I^c) = +1$ ; for  $I = \{4, 5, 6, 7\}$  the concatenation is  $(4, 5, 6, 7, 0, 1, 2, 3)$ , obtained by moving four entries past four entries, hence  $\text{sgn} = (-1)^{16} = +1$ . Then (1) gives (8). Since  $\varepsilon_1 \varepsilon_2 = (-1)^q$ , even  $q$  forces  $\varepsilon_1 = \varepsilon_2$ , and  $\varepsilon_1 = (-1)^{q_1}$  by definition. With  $\varepsilon_1 = \varepsilon_2 = \varepsilon$ ,

$$\star \Psi_{\pm} = \varepsilon \omega_2 \pm \varepsilon \omega_1 = \pm \varepsilon (\omega_1 \pm \omega_2) = \pm \varepsilon \Psi_{\pm}.$$

For odd  $q$ ,  $\star \Psi_{\pm} = \varepsilon_1 \omega_2 \pm \varepsilon_2 \omega_1 = \varepsilon_1 (\omega_2 \mp \omega_1) = \mp \varepsilon_1 \Psi_{\mp}$ , which exchanges  $\Psi_+$  and  $\Psi_-$  and therefore fixes neither.  $\square$

**Corollary 5.2** (The handedness label is a declared convention). *The eigenvalue attached to  $\Psi_+ = \omega_1 + \omega_2$  is  $\varepsilon = (-1)^{q_1}$ , so it depends on how the negative directions are distributed between the two blocks and not only on the total signature. With the negatives collected in one block,  $q_1 = 4$  and  $\Psi_+$  carries eigenvalue  $+1$ . With the block-aligned presentation  $(1, 3) \oplus (3, 1)$ , so that  $\eta = (+, -, -, -, -, +, +, +)$  and  $q_1 = 3$ , one has  $\varepsilon = -1$  and  $\Psi_+$  carries eigenvalue  $-1$ : the two labels exchange. The split  $35 \oplus 35$  and every statement of Sections 3 and 4 are unaffected. Which chirality is called positive must therefore be stated with the signature presentation and cannot be read off  $\Psi_{\pm}$  alone.*

## 6. THE CONTROL AND ITS FAILURE MODE

The value of the criterion is that it forbids something, so it must be run where it forbids. Theorem 3.2 predicts, for odd  $m^2 + q$ , not a degraded projector but a specific algebraic object:  $P_{\pm}$  with an exactly scalar idempotency deficit  $-\frac{1}{2}\text{id}$ , an exactly scalar cross-product  $+\frac{1}{2}\text{id}$ , preserved completeness  $P_+ + P_- = \text{id}$ , and full rank. Section 7 reports these four predictions measured, and the diagnostic worth naming is the rank: instead of 35 and 35 the computation returns 70 and 70. A pair of operators each of which is the whole space is the unambiguous signature of the failure, and it is visible without inspecting any entry.

Two smaller controls are also reported. Within a passing signature, the odd grades fail (Proposition 4.3(a)), so the same metric and the same star produce a chiral pair at  $k = 2$  and none at  $k = 3$ ; the signature alone therefore does not carry the result. Within a failing signature, the odd grades pass (Proposition 4.3(b)), so the star is not defective there either; only the middle grade is closed to it.

## 7. MEASURED LEDGER

All values below are produced by `chiral_projector_harness.py` in exact integer and rational arithmetic. Every entry in the matrix of  $\star$  lies in  $\{-1, 0, +1\}$  and every entry of  $P_{\pm}$  in  $\{-\frac{1}{2}, 0, \frac{1}{2}, 1\}$ ; the verdicts “is the zero matrix”, “equals +id”, “equals -id” are exact symbolic predicates and no tolerance is used anywhere. Ranks are computed over  $\mathbb{Q}$  and cross-checked by an independent floating-point singular value decomposition, which agrees.

| signature at $n = 8, k = 4$                    | $q$ | $\star\star$ | rank $P_+$ | rank $P_-$ | idem. | orth. | tr $\star$ |
|------------------------------------------------|-----|--------------|------------|------------|-------|-------|------------|
| carrier $(1, 3) \oplus (3, 1)$ , block aligned | 4   | +id          | 35         | 35         | yes   | yes   | 0          |
| $(4, 4)$ , negatives in first block            | 4   | +id          | 35         | 35         | yes   | yes   | 0          |
| $(4, 4)$ , negatives interleaved               | 4   | +id          | 35         | 35         | yes   | yes   | 0          |
| $(2, 6)$                                       | 6   | +id          | 35         | 35         | yes   | yes   | 0          |
| $(6, 2)$                                       | 2   | +id          | 35         | 35         | yes   | yes   | 0          |
| $(0, 8)$                                       | 8   | +id          | 35         | 35         | yes   | yes   | 0          |
| $(8, 0)$                                       | 0   | +id          | 35         | 35         | yes   | yes   | 0          |
| CONTROL $(1, 7)$ Lorentzian                    | 7   | -id          | 70         | 70         | no    | no    | 0          |
| CONTROL $(7, 1)$                               | 1   | -id          | 70         | 70         | no    | no    | 0          |
| CONTROL $(3, 5)$                               | 5   | -id          | 70         | 70         | no    | no    | 0          |
| CONTROL $(5, 3)$                               | 3   | -id          | 70         | 70         | no    | no    | 0          |

TABLE 1. The middle-grade ledger.  $\dim \Lambda^4(\mathbb{R}^8) = \binom{8}{4} = 70$ . The only variable that changes across the table is the parity of  $q$ ; the dimension is 8 in every row.

| quantity                                                              | measured                         | predicted                              |
|-----------------------------------------------------------------------|----------------------------------|----------------------------------------|
| $P_+^2 - P_+$ on the carrier                                          | zero matrix (exact)              | 0, Thm. 3.2                            |
| $P_-^2 - P_-$ on the carrier                                          | zero matrix (exact)              | 0                                      |
| $P_+ + P_- - \text{id}$ on the carrier                                | zero matrix (exact)              | 0                                      |
| $P_+ P_-$ on the carrier                                              | zero matrix (exact)              | 0                                      |
| rank $P_\pm$ on the carrier, over $\mathbb{Q}$                        | 35, 35                           | $\frac{1}{2} \binom{8}{4}$ , Prop. 4.1 |
| rank $P_\pm$ on the carrier, by SVD                                   | 35, 35                           | agreement check                        |
| $P_+^2 - P_+$ on control $(1, 7)$                                     | exactly $-\frac{1}{2} \text{id}$ | Eq. (4)                                |
| $P_+ P_-$ on control $(1, 7)$                                         | exactly $+\frac{1}{2} \text{id}$ | Eq. (4)                                |
| $P_+ + P_- - \text{id}$ on control $(1, 7)$                           | zero matrix (exact)              | completeness survives                  |
| rank $P_\pm$ on control $(1, 7)$                                      | 70, 70                           | full rank, Thm. 3.2                    |
| $\star$ spectrum on control $(1, 7)$                                  | purely imaginary, $\pm i$        | $\star\star = -\text{id}$              |
| $\star\omega_1, \star\omega_2$ on carrier                             | $-\omega_2, -\omega_1$           | $\varepsilon = (-1)^{q_1} = -1$        |
| $\Psi_+, \Psi_-$ eigenvalues on carrier                               | $-1, +1$                         | $\pm\varepsilon$ , Cor. 5.2            |
| $\Psi_\pm$ on control $(1, 7)$                                        | not eigenvectors                 | Prop. 5.1                              |
| $\Lambda^2 \oplus \Lambda^6$ on carrier                               | chiral pair, 28/28               | Prop. 4.3(a)                           |
| $\Lambda^3 \oplus \Lambda^5$ on carrier                               | fails, 112/112                   | Prop. 4.3(a)                           |
| $\Lambda^3 \oplus \Lambda^5$ on control $(1, 7)$                      | chiral pair, 56/56               | Prop. 4.3(b)                           |
| $\Lambda^1 \oplus \Lambda^7$ on control $(1, 7)$                      | chiral pair, 8/8                 | Prop. 4.3(b)                           |
| grade sweep $k = 0, \dots, 8$ on carrier                              | $+, -, +, -, +, -, +, -, +$      | $(-1)^{k(8-k)}$ , Lem. 3.1             |
| closed form versus computation, $2 \leq n \leq 8$ , all $k$ , all $q$ | 0 disagreements                  | Lem. 3.1                               |

TABLE 2. Detail rows. The exhaustive sweep in the last row compares  $(-1)^{k(n-k)}(-1)^q$  against the symbolically computed  $\star\star$  for every dimension  $2 \leq n \leq 8$ , every grade  $0 \leq k \leq n$ , and every  $q$  from 0 to  $n$ .

## 8. COROLLARIES AND CROSS-CHECKS

**Corollary 8.1** (The classical four-dimensional case). *At  $n = 4$ ,  $k = 2$  one has  $k(n - k) = 4$ , even, so Lemma 3.1 gives  $\star\star = (-1)^q \text{id}$  on two-forms. Hence:*

- in Euclidean signature,  $q = 0$  or  $q = 4$ , and in split signature  $(2, 2)$ , one has  $\star\star = +\text{id}$  and a real chiral pair with  $\dim \Lambda^2 = 6 = 3 \oplus 3$ ; this is the self-dual and anti-self-dual splitting;



- in Lorentzian signature,  $q = 3$  or  $q = 1$ , one has  $\star\star = -\text{id}$ , the star is a complex structure on the six-dimensional space of two-forms, and the correct object is a rotation by an angle rather than a projector.

Both branches are measured: ranks 3,3 in the even cases and 6,6 in the Lorentzian cases.

*Remark 8.2.* Corollary 8.1 is the reason the same operator appears in the series in two guises. In four dimensions the star on the middle grade is, up to sign, multiplication by the pseudoscalar, and  $\star\star = -\text{id}$  in Lorentzian signature is precisely the statement that the pseudoscalar is a complex structure. That is the operator the source current's privileged central imaginary implements, and the channel rotation  $G + S e^{i\theta}$  is then a rotation of a field into its own dual. This identification is carried from the algebraic side of the series and is not proved in this note; what this note supplies is the signature-and-grade bookkeeping that says where the same operator is a projector instead of a rotation.

*Remark 8.3* (The two twos). Corollary 8.1 and Corollary 3.3 both turn on evenness, and it is worth separating which evenness. In Corollary 8.1 the relevant parity is that of  $q$  with  $k(n - k)$  already even. In Proposition 4.3 the relevant parity is that of  $k$  with  $q$  already even. The criterion is the parity of the sum  $k(n - k) + q$ , and neither summand is the criterion by itself. Statements of the form “chirality requires even signature” and “chirality lives at even grade” are each half of (3) and are each false in the other's regime.

## 9. THE CRITERION IS THE WEYL CONDITION; THE MISSING BIT IS MAJORANA

Theorem 3.2 is a mod-4 condition on the signature invariant, and the real Clifford algebras have period 8. It would be a mistake to read the discrepancy as a mismatch of subject matter, and a worse one to read the extra factor of two as a classification label. This section identifies both halves. The criterion is the Weyl condition, exactly and not by analogy, and the invisible half is the Majorana condition. Their conjunction is the period.

### The star is the volume element.

**Lemma 9.1** (Same element, same square). *Let  $n = 2m$  and let  $\omega = e^0 \wedge \cdots \wedge e^{n-1}$  be the volume element, so that  $\omega^2 = (-1)^{n(n-1)/2}(-1)^q$  in the Clifford algebra  $\text{Cl}(p, q)$ . Then the sign of  $\star\star$  on  $\Lambda^m$  equals the sign of  $\omega^2$ , for every even  $n$  and every  $q$ .*

*Proof.* By Lemma 3.1 the exponent for  $\star\star$  at  $k = m$  is  $m^2 + q$ , and the exponent for  $\omega^2$  is  $\frac{n(n-1)}{2} + q = m(2m - 1) + q$ . Their difference is

$$m^2 - m(2m - 1) = -m^2 + m = -m(m - 1),$$

a product of consecutive integers, hence even. Two signs whose exponents differ by an even number are equal.  $\square$

*Remark 9.2.* Lemma 9.1 is the reason a computation about differential forms decides a question about spinors. Up to sign the Hodge star on the middle grade is multiplication by  $\omega$ , and  $\omega$  is also the chirality operator on the spinor module. The two live in different representations of one algebra, and the condition  $\omega^2 = +\text{id}$  is a condition on the element and not on the representation. So Theorem 3.2 is not an analogue of the chirality condition. It is the chirality condition, evaluated at grade  $m$  instead of on a spinor.



**Two invariants, and the period.****Definition 9.3** (The two Clifford signs). For a signature  $(p, q)$  with  $n = p + q$  define

$$(10) \quad w(p, q) := \operatorname{sgn}(\omega^2) = (-1)^{n(n-1)/2+q},$$

$$(11) \quad t(p, q) := \sum_I B_I^2 = \sum_{k=0}^n (-1)^{k(k-1)/2} [x^k] (1+x)^p (1-x)^q,$$

the sum in (11) running over all  $2^n$  blades  $B_I = e^{i_1} \cdots e^{i_k}$  with  $B_I^2 = (-1)^{k(k-1)/2} \varepsilon_I$ . The second equality holds because the elementary symmetric function of the  $\eta_i$  of degree  $k$  is the stated coefficient, which reduces the count from  $2^n$  terms to  $O(n^2)$  arithmetic.

**Lemma 9.4** ( $t$  is the trace form, and it reads the type). *The blades are pairwise orthogonal for the trace form on  $\operatorname{Cl}(p, q)$  with  $\langle B_I, B_I \rangle = B_I^2 \in \{\pm 1\}$ , so  $t(p, q)$  is the signature of that form. For a simple real algebra  $M_k(\mathbb{R})$  the trace form has signature  $+k$ , since it is positive on the  $\frac{1}{2}k(k+1)$ -dimensional symmetric part and negative on the  $\frac{1}{2}k(k-1)$ -dimensional antisymmetric part; for  $M_k(\mathbb{H})$  it has signature  $-2k$ , since the self-adjoint part has dimension  $k(2k-1)$  and its complement  $k(2k+1)$ ; and for  $M_k(\mathbb{C})$  it is neutral. Hence  $t > 0$ ,  $t < 0$ ,  $t = 0$  selects real, quaternionic, complex type respectively.*

At  $n = 8$  this gives  $t = +16$  for  $M_{16}(\mathbb{R})$  and  $t = -16$  for  $M_8(\mathbb{H})$ , both of which are measured.

**Proposition 9.5** (The pair is the period). *The pair  $(w(p, q), \operatorname{type}(p, q))$  depends only on  $p - q$  modulo 8, according to Table 3. Measured on 189 signatures with  $n \leq 18$  with zero conflicts.*

| $p - q \bmod 8$ | $w$ | type         | Weyl | Majorana | Majorana–Weyl | reading               |
|-----------------|-----|--------------|------|----------|---------------|-----------------------|
| 0               | +1  | $\mathbb{R}$ | yes  | yes      | <b>yes</b>    | splits, real          |
| 1               | +1  | $\mathbb{R}$ | yes  | yes      | <b>yes</b>    | splits, real          |
| 2               | −1  | $\mathbb{R}$ | no   | yes      | no            | complex, real         |
| 3               | −1  | $\mathbb{C}$ | no   | no       | no            | complex type          |
| 4               | +1  | $\mathbb{H}$ | yes  | no       | no            | splits, quaternionic  |
| 5               | +1  | $\mathbb{H}$ | yes  | no       | no            | splits, quaternionic  |
| 6               | −1  | $\mathbb{H}$ | no   | no       | no            | complex, quaternionic |
| 7               | −1  | $\mathbb{C}$ | no   | no       | no            | complex type          |

TABLE 3. The two signs as functions of the signature invariant modulo eight. Column  $w$  is what Theorem 3.2 sees. The type column is what it cannot see.

**Definition 9.6** (The two spinor conditions). Call the signature *Weyl* if  $w = +1$ , so that  $\omega$  is a central involution of the even subalgebra, the even subalgebra splits as a direct sum of two ideals, and the two half-spin modules are inequivalent over the reals. Call it *Majorana* if the type is real, so that the spinor module admits a real structure. Call it *Majorana–Weyl* if both.

**Corollary 9.7** (The period is the conjunction). *The Weyl condition is  $p - q \equiv 0 \pmod{4}$  and the Majorana condition is  $p - q \equiv 0, 1, 2 \pmod{8}$ . Their conjunction is  $p - q \equiv 0, 1 \pmod{8}$ , and since even  $n$  forces  $p - q$  even, for even  $n$  the Majorana–Weyl condition is exactly*

$$(12) \quad p - q \equiv 0 \pmod{8}.$$

*Proof.* Read the two columns of Table 3 and intersect. The reduction of  $w = +1$  to  $p - q \equiv 0 \pmod{4}$  is direct from (10): writing  $n = 2a$  and  $d = p - q = 2b$ , the exponent is  $a(2a-1) + q$  and  $q = a - b$ , so the exponent is  $2a^2 - b$ , whose parity is that of  $b = d/2$ ; hence  $w = +1$  if and only if  $d \equiv 0 \pmod{4}$ . Measured for all even  $n \leq 16$  and all  $q$  with zero mismatches.  $\square$

*Remark 9.8* (Anchors). Equation (12) is the standard condition and the harness recovers its familiar instances. In Lorentzian signature  $(1, d-1)$  one has  $p-q = 2-d$ , so Majorana–Weyl requires  $d \equiv 2 \pmod{8}$ , that is  $d = 2, 10, 18$ ; the case  $d = 10$  is measured, as are the failures at  $d = 4$  and  $d = 11$ . Split signature  $(m, m)$  has  $p-q = 0$  and is Majorana–Weyl in every even dimension.

*Remark 9.9* (A convention that must be declared).  $\text{Cl}(1, 3)$  has  $p-q = -2 \equiv 6$  and is of quaternionic type;  $\text{Cl}(3, 1)$  has  $p-q = +2$  and is of real type. Both are measured. So the four-dimensional Majorana statement inverts with the sign convention chosen for the metric, and any assertion about the existence of a Majorana condition must carry its convention in the same sentence. The reading with  $\text{Cl}(3, 1)$  is the textbook one: Majorana yes, Weyl no, Majorana–Weyl no.

### The forcing.

**Theorem 9.10** (The signature is forced in eight dimensions). *Impose on a signature  $(p, q)$  with  $p+q = 8$  the three requirements*

- (i) *real chirality at the middle grade, that is  $w = +1$ , equivalently the chiral pair of Theorem 3.2 exists;*
- (ii) *a real structure on the spinor module, that is real type;*
- (iii) *a nonempty light cone, that is  $p > 0$  and  $q > 0$ .*

*Then  $(p, q) = (4, 4)$ , uniquely.*

*Proof.* By Corollary 9.7, (i) and (ii) together are  $p-q \equiv 0 \pmod{8}$ . With  $p+q = 8$  the invariant  $p-q$  takes the values  $8, 6, 4, 2, 0, -2, -4, -6, -8$  as  $q$  runs from 0 to 8, of which the multiples of eight are  $8, 0$  and  $-8$ , that is  $q \in \{0, 4, 8\}$ . Requirement (iii) excludes  $q = 0$  and  $q = 8$ . The survivor is  $q = 4$ . Measured: Table 4.  $\square$

| $q$      | $p-q$    | mod8     | type         | (i) Weyl   | (ii) Majorana | (iii) indefinite |
|----------|----------|----------|--------------|------------|---------------|------------------|
| 0        | +8       | 0        | $\mathbb{R}$ | yes        | yes           | <b>no</b>        |
| 1        | +6       | 6        | $\mathbb{H}$ | no         | no            | yes              |
| 2        | +4       | 4        | $\mathbb{H}$ | yes        | <b>no</b>     | yes              |
| 3        | +2       | 2        | $\mathbb{R}$ | <b>no</b>  | yes           | yes              |
| <b>4</b> | <b>0</b> | <b>0</b> | $\mathbb{R}$ | <b>yes</b> | <b>yes</b>    | <b>yes</b>       |
| 5        | -2       | 6        | $\mathbb{H}$ | no         | no            | yes              |
| 6        | -4       | 4        | $\mathbb{H}$ | yes        | <b>no</b>     | yes              |
| 7        | -6       | 2        | $\mathbb{R}$ | <b>no</b>  | yes           | yes              |
| 8        | -8       | 0        | $\mathbb{R}$ | yes        | yes           | <b>no</b>        |

TABLE 4. The forcing of Theorem 9.10, measured. Exactly one row satisfies all three requirements. Bold entries are the failing requirement in each excluded row.

*Remark 9.11* (What this closes). The previous draft recorded as open the question of what distinguishes  $(4, 4)$  from the other signatures passing the criterion, remarking that the criterion selects a parity class and not a signature. That remark is correct about the criterion and wrong about the conclusion. The quaternionic candidates  $(6, 2)$  and  $(2, 6)$  pass requirement (i) and fail (ii): they carry a symplectic rather than a genuine Majorana condition, so a chiral field on them cannot be its own conjugate. The definite candidates  $(8, 0)$  and  $(0, 8)$  pass (i) and (ii) and fail (iii): their light cone is empty, so the past-cone integration on which the accumulation law is defined has empty domain. What separates  $(4, 4)$  from  $(2, 6)$  is therefore a physical requirement on the field content and not a shelf in a classification.

*Remark 9.12* (What this does not close). Note that  $\text{Cl}(4, 4) \cong \text{Cl}(8, 0) \cong M_{16}(\mathbb{R})$ , both sitting at  $p - q \equiv 0$ , so no invariant of the algebra separates them; requirement (iii) is metric data and not algebraic data, and it is doing real work in Theorem 9.10.

More importantly, the forcing selects a signature *given* a dimension and does not select the dimension. Measured:  $n = 2, 4, 6, 8$  each admit exactly one indefinite Majorana–Weyl signature, namely the split one, while  $n = 10, 12, 14, 16$  admit three and  $n = 18, 20, 22, 24$  admit five. So eight is not distinguished by these three requirements. Eight comes from the division algebra tower terminating there, which is Adams’ theorem and which rests on the periodicity of Section 9 rather than on anything proved here. The correct statement is a chain of two links and should be written as one: the period fixes the dimension through the tower, and the period fixes the signature within that dimension through Theorem 9.10. Presenting the second link as though it supplied the first would be exactly the overreach that a mod-eight condition invites.

*Remark 9.13* (The two halves and triality). One dimension count is worth recording against Remark 10.3. The two halves of Proposition 4.1 have dimension 35 each, and 35 is also  $\dim \text{Sym}_0^2(\mathbb{R}^8) = 36 - 1$ . That is not a coincidence of counting: in eight dimensions the rotation group carries an outer automorphism of order three, and triality carries the two middle-grade half-spaces to the traceless symmetric squares of the vector representation. Under that identification the two halves are irreducible and are the pair triality acts on. This is *carried* and not proved here; what is measured is only the dimension agreement.

## 10. REMARKS ON SCOPE

*Remark 10.1* (What is discharged). The series stated that the carrier admits a native chiral projector and that the even negative count is what permits it. Theorem 3.2 and Corollary 3.3 prove it, with the necessary direction as well as the sufficient one. Proposition 4.1 supplies the dimension count that the assertion left open, and Proposition 4.3 supplies the grade condition that the assertion did not contain.

*Remark 10.2* (What is corrected). Corollary 5.2 corrects a label. Writing  $\Psi_{\pm} = \omega_1 \pm \omega_2$  and calling  $\Psi_+$  the positive chirality is consistent only when the negative directions are collected in one block. In the block-aligned presentation  $(1, 3) \oplus (3, 1)$  the eigenvalues attached to  $\Psi_+$  and  $\Psi_-$  exchange. This is a convention and not an error, but it must be declared, because the two presentations differ by nothing observable and give opposite handedness names.

*Remark 10.3* (Open). Two items are open and are not addressed here. First, this note produces a real splitting of  $\Lambda^4$  into two 35-dimensional halves; whether those halves carry the representation content the framework requires of a physical chirality, as against merely having the right dimension, is a question about the action of the relevant structure group on the eigenspaces, and Remark 9.13 records only the dimension agreement and the triality identification as carried. Second, the extension of the criterion to the framework’s helical density operators, which carry paired indices rather than a single multi-index, follows from Proposition 4.3 only once those operators are exhibited as elements of a fixed even grade; that exhibition is not performed here.

The item on which signature is selected, recorded as open in the previous draft, is closed by Theorem 9.10, and the residual honest limitation is stated separately in Remark 9.12: the forcing fixes a signature within a dimension and does not fix the dimension.

*Remark 10.4* (One sentence). The star squares to the sign of the determinant times  $(-1)^{k(n-k)}$ , so a projector exists exactly when those two parities agree; at the middle grade of an eight-dimensional carrier the grade parity is automatic and the whole condition is the parity of the negative count, which is why the split is 35 and 35 and why the same construction in Lorentzian eight dimensions returns two operators of full rank and an idempotency deficit of exactly one half;

and because that condition is a condition on the volume element rather than on the module it acts in, it is the Weyl condition, so it is mod four, and the mod-eight period of the real Clifford algebras is what one gets by conjoining it with the Majorana condition, which is why requiring a real chirality and a light cone at once leaves  $(4, 4)$  alone in eight dimensions.

## APPENDIX A. NOTATION

|                         |                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------|
| $n, k, m$               | carrier dimension; form degree; the middle grade $m = n/2$ .                                      |
| $(p, q)$                | signature; $q$ is the number of negative directions, $\text{sgn det } \eta = (-1)^q$ .            |
| $q_1$                   | number of negative directions among indices $0, 1, 2, 3$ ; fixes the handedness label (Cor. 5.2). |
| $\eta_i, \varepsilon_I$ | diagonal entries of the form; $\varepsilon_I = \prod_{i \in I} \eta_i$ .                          |
| $I, I^c$                | sorted multi-index and its sorted complement.                                                     |
| $\text{sgn}(I, I^c)$    | sign of the permutation carrying $(0, \dots, n-1)$ to $(I, I^c)$ .                                |
| $\star$                 | Hodge star of Convention 2.2; $\Lambda^k \rightarrow \Lambda^{n-k}$ .                             |
| $s$                     | the scalar of Lemma 3.1, $s = (-1)^{k(n-k)+q}$ .                                                  |
| $P_\pm$                 | $\frac{1}{2}(\text{id} \pm \star)$ on $\Lambda^m$ (Def. 2.3).                                     |
| $Z_k, P_\pm^{(k)}$      | paired-grade operator and its projectors (Def. 2.4).                                              |
| $\omega_1, \omega_2$    | block volume forms (7).                                                                           |
| $\Psi_\pm$              | $\omega_1 \pm \omega_2$ ; eigenvectors with eigenvalue $\pm \varepsilon$ .                        |
| $\varepsilon$           | $(-1)^{q_1}$ , the block sign of Prop. 5.1.                                                       |
| $\omega$                | Clifford volume element $e^0 \dots e^{n-1}$ ; the chirality operator.                             |
| $w(p, q), t(p, q)$      | the two Clifford signs (10), (11): sign of $\omega^2$ , and the trace-form signature.             |
| $B_I$                   | the blade $e^{i_1} \dots e^{i_k}$ ; $B_I^2 = (-1)^{k(k-1)/2} \varepsilon_I$ .                     |
| type                    | real, complex or quaternionic, read from $\text{sgn } t$ (Lem. 9.4).                              |

## APPENDIX B. HARNESS

`chiral_projector_harness.py` is distributed with this note. It constructs  $\star$  from (1) directly, in exact integer arithmetic, with no library implementation of the Hodge star and no metric-dependent shortcut, and then evaluates: the star-square verdict against the closed form of Lemma 3.1; the four projector predicates of Definition 2.3 as exact zero-matrix tests; the exact rational ranks; the exact scalar values of the failure constants (4); the block eigenvectors of Proposition 5.1 with their predicted  $\varepsilon = (-1)^{q_1}$ ; the paired-grade sweep of Proposition 4.3; the grade sweep  $k = 0, \dots, n$ ; an exhaustive comparison of the closed form against the computed matrix for all  $2 \leq n \leq 8$ , all  $k$ , and all  $q$ ; and, on the Clifford side of Section 9, the identity of Lemma 9.1 on 168 signatures with  $n \leq 24$ , the generating-function form (11) against the exponential blade sum for all  $n \leq 12$ , the type discriminator against the seven anchors of Lemma 9.4, the mod-8 dependence of Proposition 9.5 on 189 signatures with  $n \leq 18$ , the spinor anchors, and the forcing of Theorem 9.10 together with its dimension sweep. Invocation:

```
python3 chiral_projector_harness.py      (full ledger)
python3 chiral_projector_harness.py --quick (carrier and primary control)
```

The script exits with status equal to the number of disagreements between the closed form and the computation, so it is usable directly as a regression test. On the reported run that number is zero.

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LIOR RESEARCH INITIATIVE

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